

CommandIA

Final Task Division & 4-Week Execution Roadmap

Project Name	CommandIA
Project Scope	Conversational AI Order Automation for Algerian E-Commerce
Target Deliverable	Working AI System (Instagram DM → Webhook → Supabase) + Clickable Figma Merchant Dashboard + Final Demo Video Pitch
Sprint Duration	4 Weeks (~7-10 hrs/week per role)

Team Role Overview

Role 1 Project Lead & Backend Infrastructure Engineer	Primary Focus: Project coordination, Meta API integration, FastAPI webhooks, Supabase database management, system integration, pitch narration.
Role 2 AI / NLP & Data Engineering Specialist	Primary Focus: Social media data scraping/curation, "Golden Dataset" development, prompt engineering for Algerian Darija/Franco-Arabic, Pydantic JSON schema enforcement, technical demo recording.
Role 3 UI/UX & Media Design Specialist	Primary Focus: Competitor benchmark research, low/high-fidelity Figma wireframes, interactive dashboard prototyping, pitch deck slides, video editing and final assembly.

4-Week Execution Plan

Week 1: Infrastructure, Data Curation & Wireframing

Role 1 (Project Lead & Backend Infrastructure Engineer)

- Create and verify the Meta Developer Account and App container (CommandIA-Test).
- Set up a test Facebook Business Page and linked Instagram Professional Account.
- Initialize the backend project repository using Python (FastAPI).
- Design and deploy the Supabase PostgreSQL database schema (Merchants, Products, Orders).

Role 2 (AI / NLP & Data Engineering Specialist)

- Execute comment/DM scraping scripts across public Algerian e-commerce social posts using instaloader or web scraping tools.
- Curate a 50-entry "Golden Dataset" in Google Sheets mapping raw customer inputs (in Darija, Franco-Arabic, French, and English) to expected JSON outputs.
- Draft the V1 System Prompt for the LLM engine defining dialect rules and commercial guardrails (e.g., no unauthorized discounts).

Role 3 (UI/UX & Media Design Specialist)

- Conduct benchmark UX research on local and international merchant portals.
- Design low-fidelity wireframes in Figma for the core dashboard views:
 - Incoming Orders Table (Filtering by status, Wilaya, and delivery type).
 - Product Catalog Manager (Adding items, pricing, inventory).
 - Bot Settings (Toggling AI ON/OFF, setting delivery fees).

Week 2: Pipeline Development & High-Fidelity Design

Role 1 (Project Lead & Backend Infrastructure Engineer)

- Build the FastAPI webhook endpoint to verify and receive live Meta Graph API webhook payloads (GET/POST).
- Implement local tunneling (ngrok) to test live incoming DMs from Instagram.
- Build the Supabase database connector to write structured JSON order payloads directly to the Orders table.

Role 2 (AI / NLP & Data Engineering Specialist)

- Implement the LLM engine integration using Python (OpenAI or Anthropic APIs).
- Enforce strict JSON output parsing using Pydantic or Instructor for the 5 Sacred Fields:
 - customer_name
 - phone_number (Validated 10-digit Algerian format)
 - wilaya (Official 58 Wilayas)
 - delivery_type (HOME vs. DESK)
 - address (Commune / Street / Bureau details)
- Test prompt iterations against the "Golden Dataset" to refine handling of haggling and missing fields.

Role 3 (UI/UX & Media Design Specialist)

- Convert low-fidelity wireframes into high-fidelity Figma visual designs.
- Integrate custom branding, Algerian localizations (Wilaya/Commune selectors, delivery type badges), and dark/light UI components.
- Design visual components for the 5 Sacred Fields display within the orders management view.

Week 3: End-to-End System Integration & Pitch Preparation

Role 1 (Project Lead & Backend Infrastructure Engineer)

- Connect the FastAPI Webhook pipeline directly to the AI Engine and Supabase database.
- Conduct end-to-end integration testing: send live DMs from test Instagram accounts and confirm automated DB records.
- Implement exception handling for edge cases (unsupported formats, API drop-outs, incomplete data).

Role 2 (AI / NLP & Data Engineering Specialist)

- Fine-tune the conversational state machine logic (State 0: Inquiry → State 1: Intent → State 2: Data Collection → State 3: Final Confirmation).
- Implement human handoff triggers (flagging chats requiring manual intervention in the database).
- Prepare the technical demonstration environment for video capture.

Role 3 (UI/UX & Media Design Specialist)

- Connect high-fidelity Figma screens into an interactive, clickable prototype simulating a live merchant session.
- Design 3-4 presentation slides for the pitch video (Problem, Solution, Architecture, Market Opportunity).
- Draft the video storyboard and visual script in coordination with the team.

Week 4: Final Demo Production & Project Submission

Role 1 (Project Lead & Backend Infrastructure Engineer)

- Coordinate team final deliverables and submission compliance.
- Record the project overview, problem statement, and business value voiceover audio track.
- Perform final code review and push backend repository documentation to GitHub.

Role 2 (AI / NLP & Data Engineering Specialist)

- Record the split-screen technical demonstration:
 - **Left Screen:** Live Instagram chat interaction in Darija/Franco-Arabic on mobile.
 - **Right Screen:** Real-time JSON data extraction and automated row creation in Supabase.

Role 3 (UI/UX & Media Design Specialist)

- Record screen walkthroughs of the clickable Figma prototype.
- Edit the final submission video (combining voiceover narration, technical split-screen demo, dashboard walkthrough, and pitch slides).
- Add captions, audio levelling, and export the final high-definition demo video file.

System Interface & Data Contract

To ensure independent development between Backend, AI, and Design, all components must align on this exact JSON schema:

```
{
  "order_id": "uuid-v4",
  "customer_name": "String",
  "phone_number": "String (10 Digits: 05/06/07...)",
  "wilaya": "String (1-58)",
  "delivery_type": "Enum ('HOME', 'DESK')",
  "address": "String (Commune / Location Details)",
  "product_details": {
    "item_id": "String",
    "quantity": "Integer",
    "size_color": "String"
  },
  "total_price_da": "Float",
  "order_status": "Enum ('PENDING_CONFIRMATION', 'CONFIRMED', 'REQUIRES_HUMAN')"
}
```

Daily Operational Protocol

Daily Asynchronous Standup: Every team member posts three brief updates in the central project chat at the end of the day:

- **Progress:** What was completed today.
- **Next Steps:** What will be worked on tomorrow.
- **Blockers:** Any technical issues or dependencies needing team support.